

Set up your Databricks Workspace

The event will begin at 9:05 AM Pacific

<https://bit.ly/420fd9c>

Step 0

- Copy the bit.ly link into an **incognito window**; **Note: browser must be in incognito mode to access the workspace for the lab**

Step 1

- See Workspace Registration: The temporary workspace you'll be using in today's workshop is administered by a third party vendor
- Complete this short registration form for the hands-on portion of the workshop

Step 2

- Launching the workspace: Click "Launch Lab" in the email you receive from Cloudlabs and allow 5-10 minutes for the environment to load

Step 3

- Accessing the Workspace: After the environment has been prepared, copy-paste the provided workspace link into an **incognito window**
- **Note: Using an incognito window will prevent you from logging into your company's workspace. If issues persists, try disconnecting from VPN and trying again**

Step 4

- Signing in on **AWS**: Sign in with the provided username and password from the credentials page

Step 5

- Arrive in your Databricks Workspace!



Data Intelligence with Databricks on AWS

powered by aws



Agenda

Data Intelligence with Databricks on AWS Workshop

9:00-9:20 AM	Databricks Keynote
9:20-9:30 AM	AWS Presentation
9:30-10:30 AM	Data Engineering Hands-on Lab
10:30-11:10 AM	Data Science and Machine Learning Hands-on Lab
11:10-11:50 AM	Databricks SQL Hands-on Lab
11:50-NOON	Q&A

Speakers



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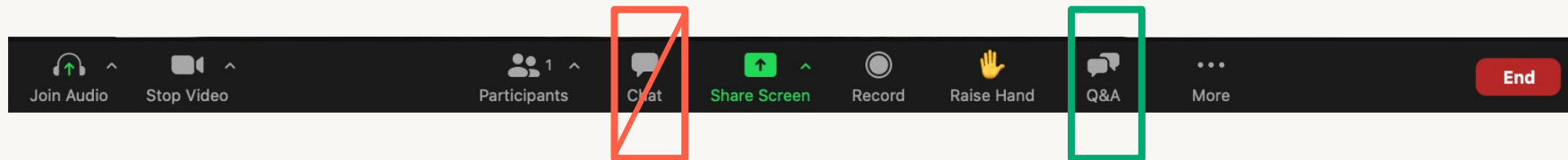


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Housekeeping Details



- The chat will be disabled. We ask that you please post all questions in the **Q&A**. We have a team monitoring this, so we can answer your questions throughout the event.
- We will also save time at the end to address any new or outstanding questions live.
- The recording and slides from today will be sent to you via email later this week.
- The lab environment you used today will be available until 10 PM PT on March 27th



Democratizing Data + AI

Sue Ewig
Databricks

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12,000+
global customers

Inventor of the **lakehouse**
&
Pioneer of **generative AI**

\$1.6B+
in revenue



databricks
The data and AI company

\$4B
in investment

Gartner-recognized Leader
Database Management Systems
Data Science and Machine Learning Platforms

Creator of



DELTA LAKE

mlflow™

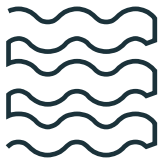
APACHE
Spark™



The winners in every industry will be
data + AI companies



Data Lake



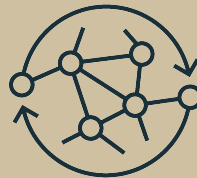
Machine Learning



Streaming



Generative AI



Data Science



Most organizations struggle
to realize this vision

Governance



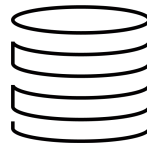
Orchestration
& ETL



BI



Data Warehouse



Data Lake



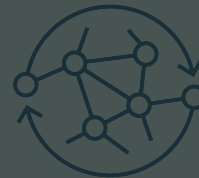
Machine Learning



Streaming



Generative AI



Data, AI, and
governance are siloed

Data privacy & control
are challenged by AI

Dependent on highly
technical staff

Data Science



Governance



Orchestration
& ETL



BI

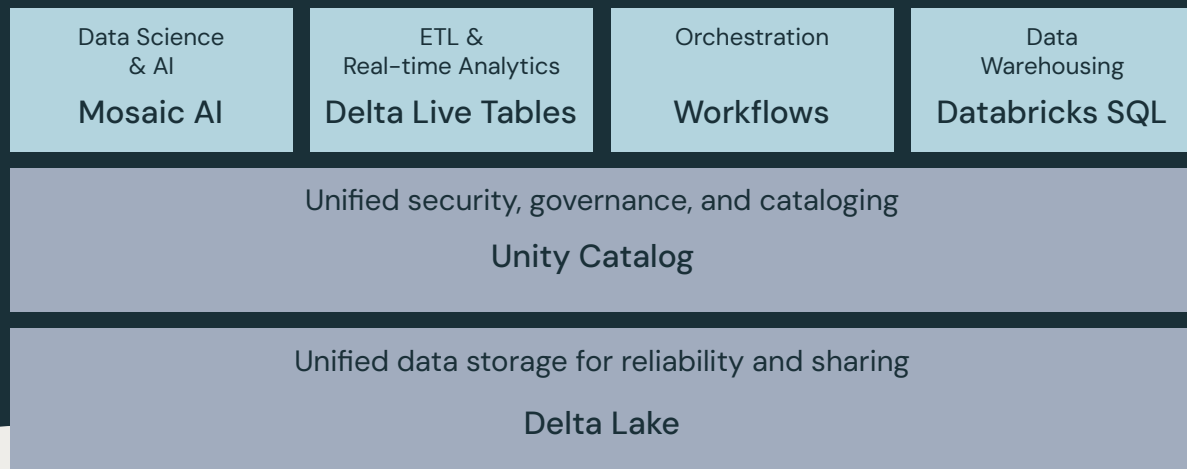


Warehouse



The Data Lakehouse

An open, unified foundation for all your data



Open Data Lake

All Raw Data
(Logs, Texts, Audio, Video, Images)

2020

Databricks pioneered the lakehouse architecture

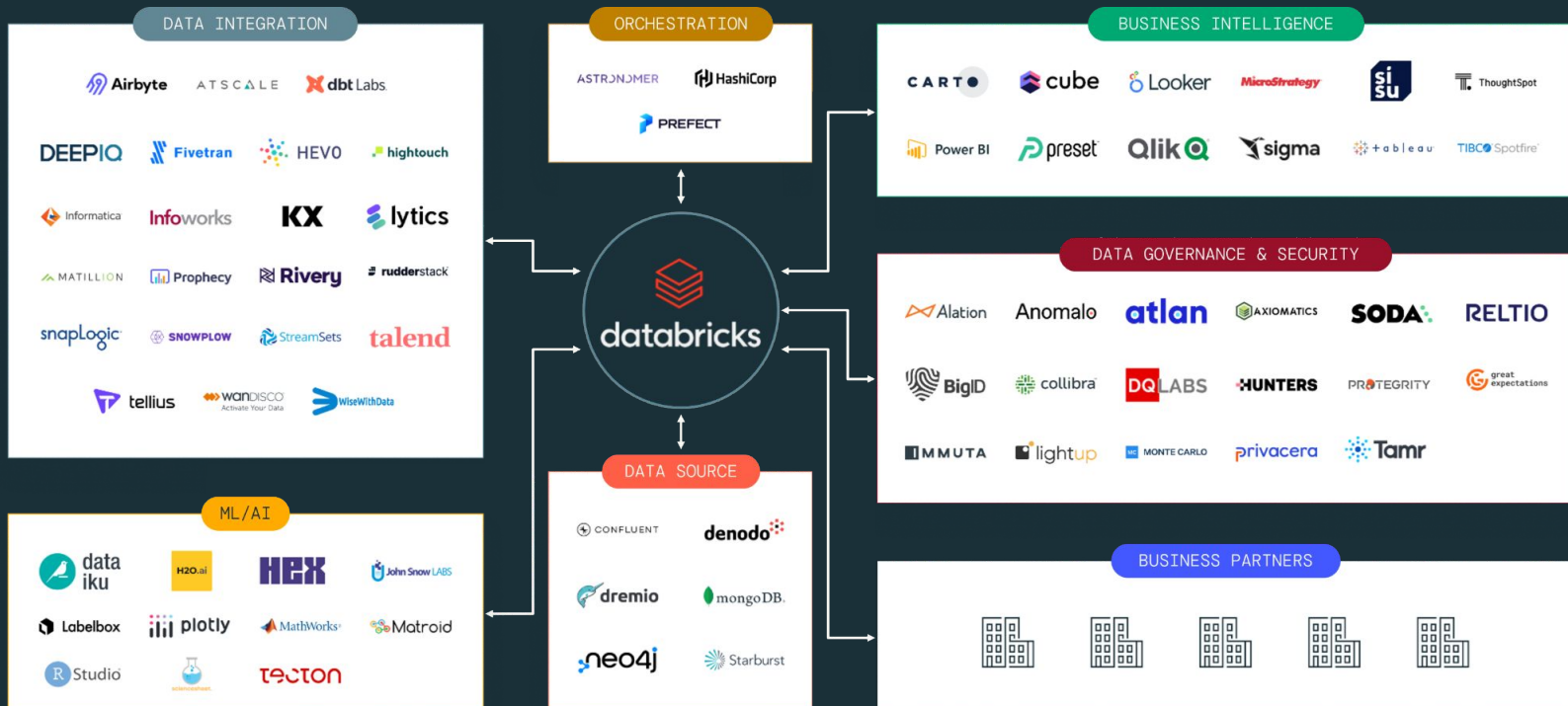
Today

74% of global enterprises have adopted lakehouse

MIT Technology Review Insights, 2023

Built on an open foundation

Easily integrate with the entire data and AI ecosystem



Data Lakehouse

An open, unified foundation
for all your data



Generative AI

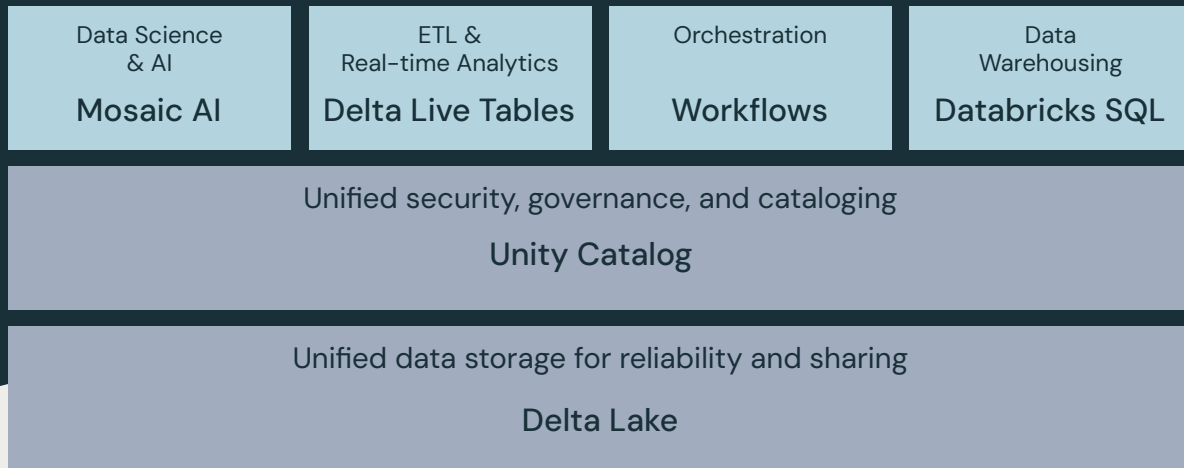
Easily scale and use data and AI



Data Intelligence Platform

Democratize data + AI across
your entire organization

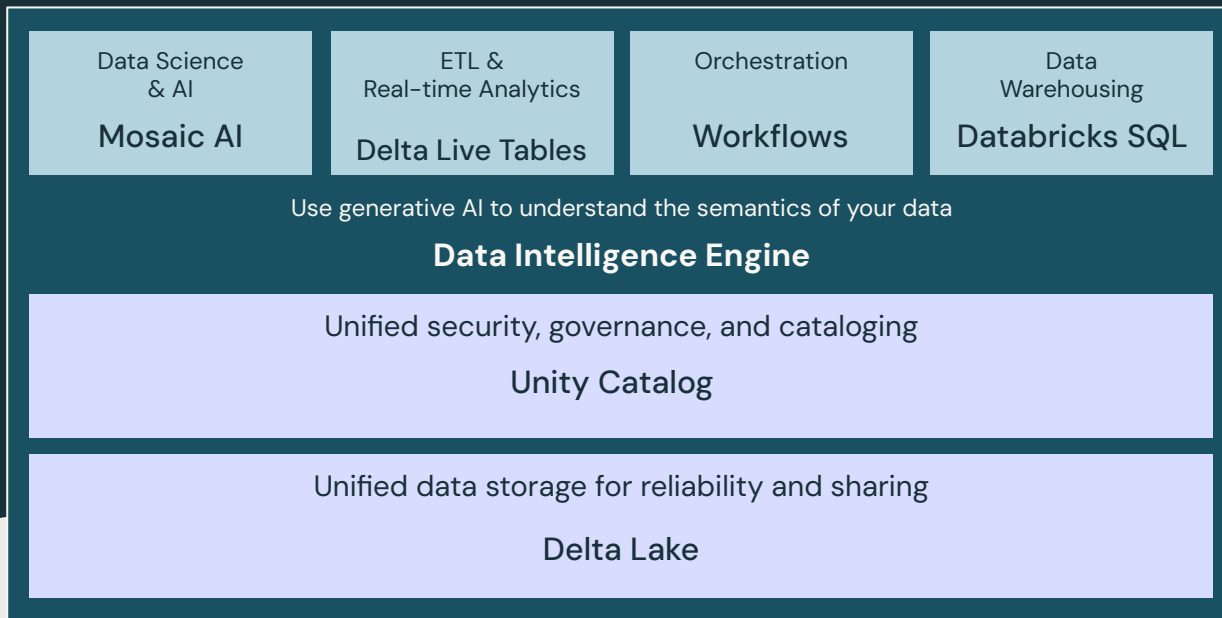
Databricks Data Intelligence Platform



Open Data Lake

All Raw Data
(Logs, Texts, Audio, Video, Images)

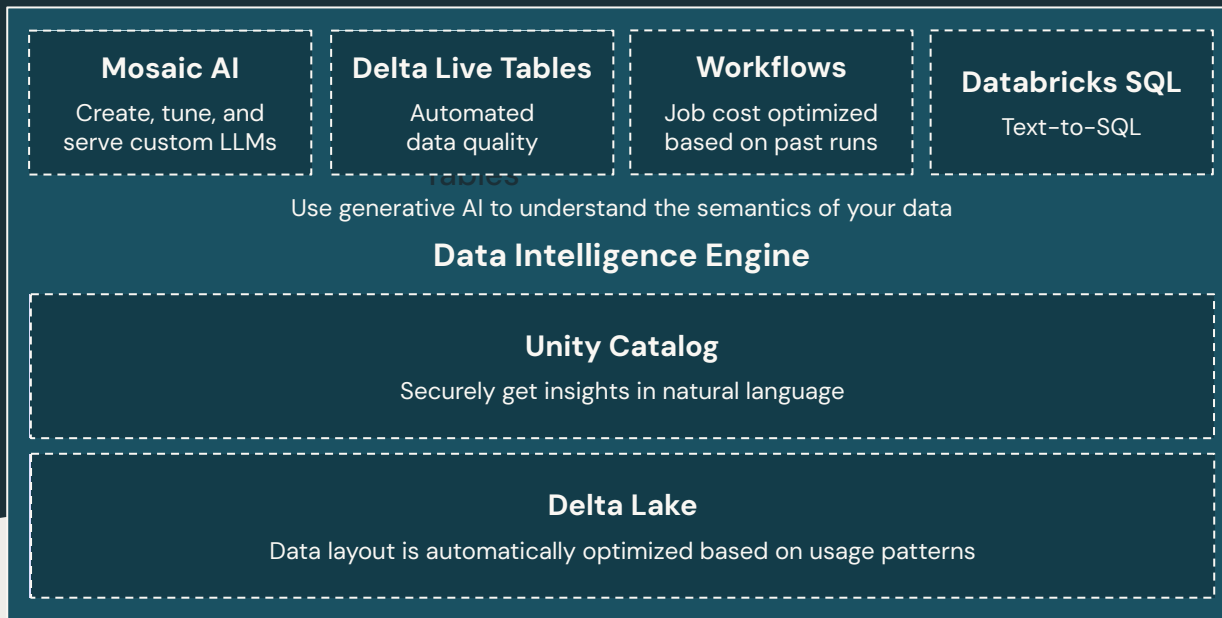
Databricks Data Intelligence Platform



Open Data Lake

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Databricks Data Intelligence Platform



Open Data Lake

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Databricks Data Intelligence Platform

Mosaic AI

GenAI

- Fine-tuned models
- Pre-training models
- Model serving
- RAG
- VectorDB

Production AI

- MLOps/MLflow
- Feature Store
- AutoML
- Monitoring
- Model Governance

Mosaic AI
Create, tune and
serve custom LLMs

Delta Live Tables
Automated
data quality

Workflows
Job cost optimized
based on past runs

Databricks SQL
Text-to-SQL
Text-to-Viz

Use generative AI to understand the semantics of your data

Data Intelligence Engine

Unity Catalog

Securely get insights in natural language

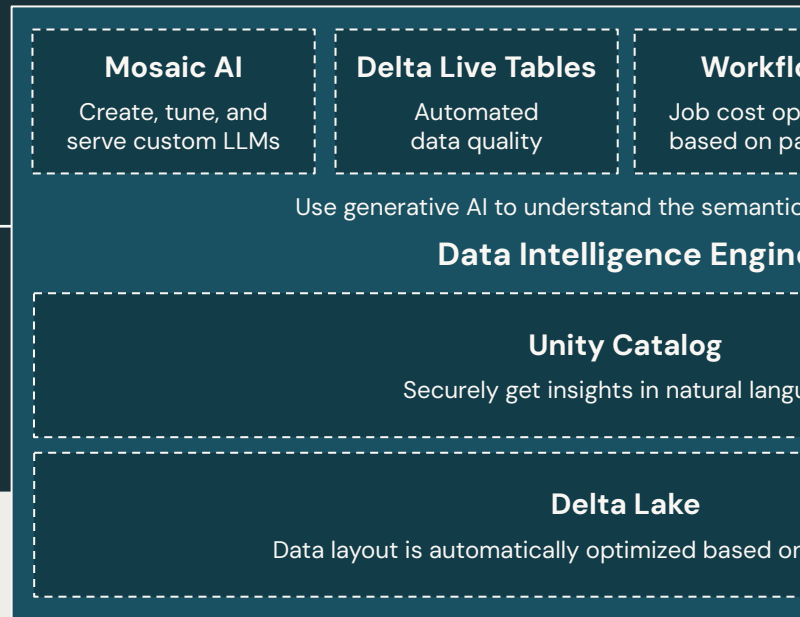
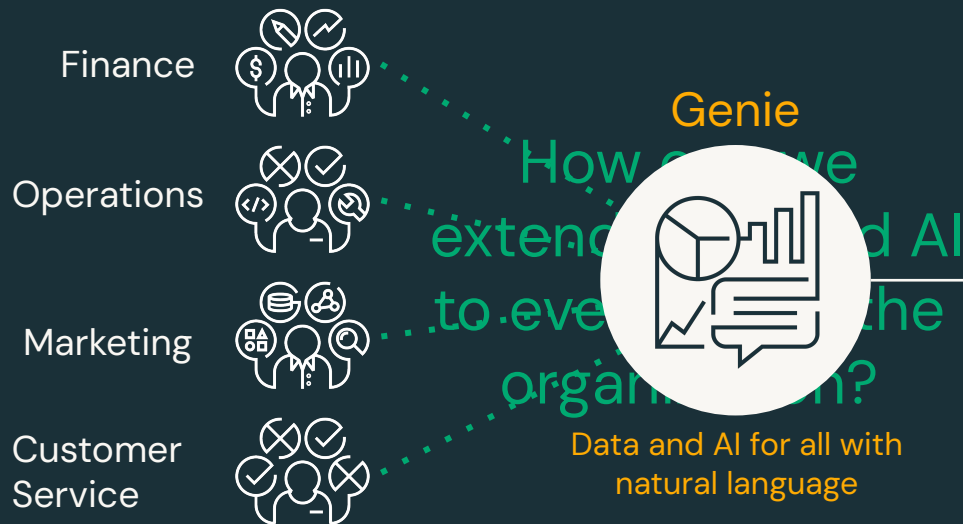
Delta Lake

Data layout is automatically optimized based on usage patterns

Open Data Lake

All Raw Data
(Logs, Texts, Audio, Video, Images)

Databricks Data Intelligence Platform



**Data
Lakehouse**



**Generative
AI**



**Data
Intelligence**

Democratize **Data + AI**

ACCESSIBLE TO EVERYONE

Natural language provides
ease of use and
efficiency for all

SIMPLE TO OPERATE

AI models used to tune
and remove almost all
knobs
in the platform

BUILD YOUR OWN PRIVATE AI

Custom AI models are easily
built on your private data





Databricks on AWS

The Data Intelligence Platform
seamlessly integrated with AWS



What sets AWS apart?



Security and Compliance

More security standards and compliance certifications than any other cloud



Breadth & Depth; Pace of Innovation

200+ fully featured services to support any cloud workload; rapid customer driven releases



Global Reach and High Availability

105 Availability Zones within 36 geographic Regions, 42 Local Zones, 29 Wavelength Zones, 600+ [Points of Presence](#) (400+ Edge locations and 13 regional edge caches).



Improve TCO

134 price reductions since 2006



Machine Learning

AWS offers the broadest and deepest set of machine learning services and supporting cloud infrastructure, putting machine learning in the hands of every developer, data scientist and expert practitioner.



Largest community of customers and partners

AWS has millions of active customers every month. The AWS Partner Network has more than 130,000 Partners from more than 200 countries, with almost 70% headquartered outside of the United States.



Enterprise leader

AWS is the longest-running Magic Quadrant™ Leader

Security and Compliance

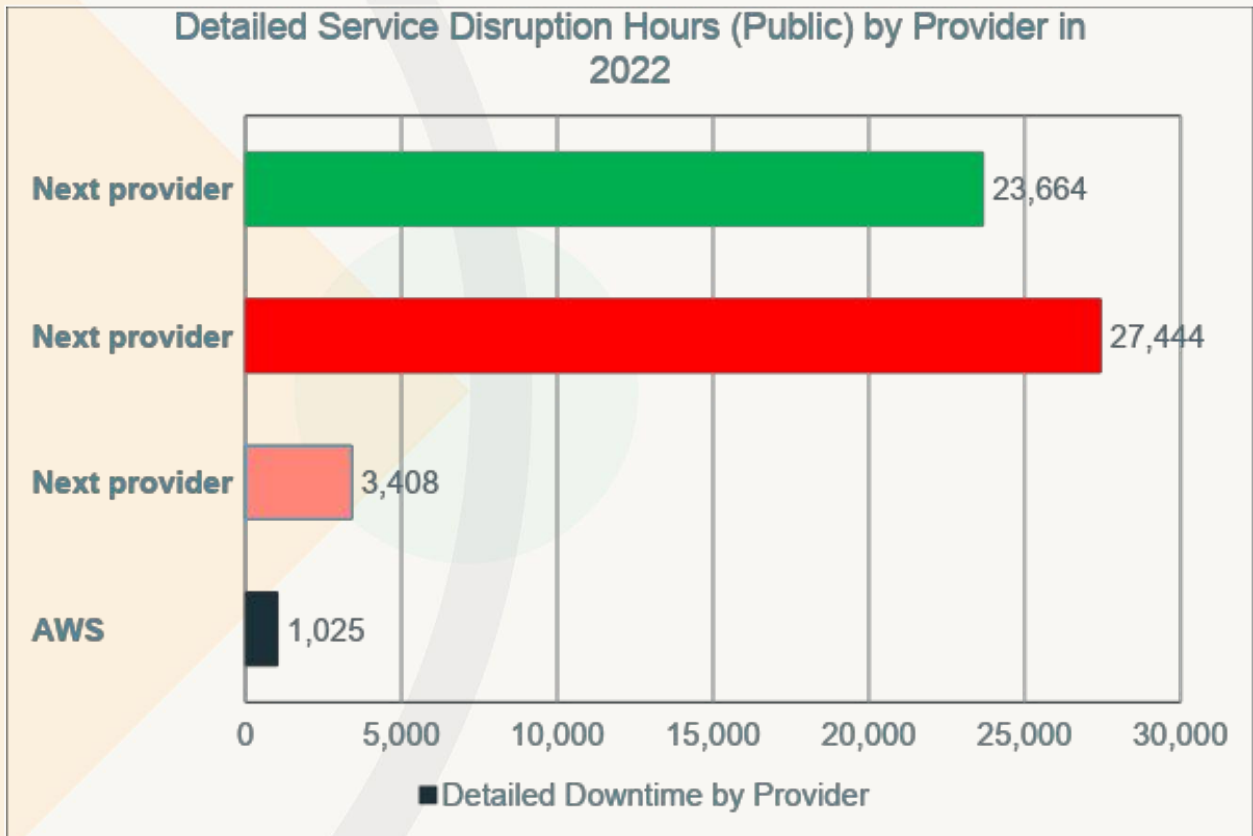


"AWS allowed us to store information in a cost effective manner while alleviating the burden of supporting the necessary infrastructure since AWS takes care of that. It really is a win-win for us and our customers."

AWS supports more security standards and compliance certifications than any other offering, including PCI-DSS, HIPAA/HITECH, FedRAMP, GDPR, FIPS 140-2, and NIST 800-171, helping satisfy compliance requirements for virtually every regulatory agency around the globe.

Service resiliency

Service disruption hours published = Sum ((Service disruption hours per incident and product) x regions impacted)



* Source: Public Status Pages for Cloud Providers

Generative AI Stack

APPLICATIONS THAT LEVERAGE LLMs AND OTHER FMs



Amazon Q
Business



Amazon Q
Developer



Amazon Q in
QuickSight



Amazon Q in
Connect

TOOLS TO BUILD WITH LLMs AND OTHER FMs



Amazon Bedrock

Guardrails | Agents | Customization Capabilities

INFRASTRUCTURE FOR FM TRAINING AND INFERENCE



GPUs



Trainium



Inferentia



SageMaker



UltraClusters



EFA



EC2 Capacity
Blocks



Nitro



Neuron



Amazon Bedrock

Helps keep your data secure
and private

None of the customer's data is
used to train the underlying model

All data is encrypted in transit and
at rest; data used for customization
is securely transferred through
customer's VPC

Data remains in the AWS Region
where the aPI is processed

Support for GDPR, SOC, ISO, CSA
compliance, and HIPAA eligibility

Cross-region inference

Dynamically route requests
across regions to manage
traffic burts



Using on-demand mode to get higher throughput limits (up to 2x of in-region quota) for optimal availability, performance, and resiliency during periods of peak demand



Prioritizing the region of the connected Bedrock API endpoint whenever possible, and minimizing latency to a nearby opted-in region



Selecting either a US Model or EU Model from 2–3 geographic regions of Claude 3 and 3.5 Sonnet models today, more regions and models support coming soon



All traffic remains within the AWS Backbone network between regions for secure and reliable execution



No additional routing or data transfer cost and you pay the same price per token for models as in your primary region

Databricks and AWS better together



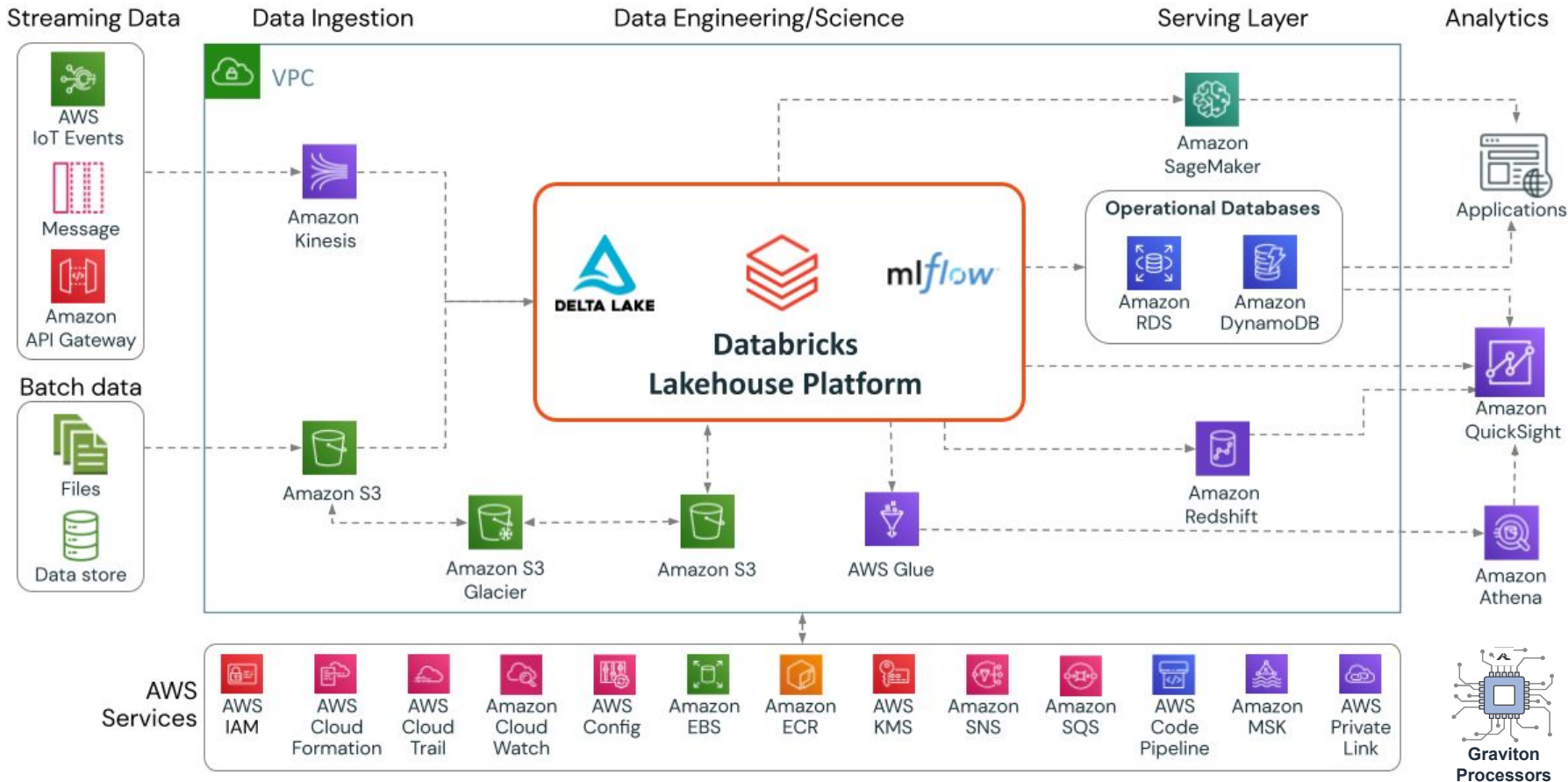
AWS Competencies:

- Advertising and Marketing Technology Software Competency
- Data & Analytics ISV Competency
- Energy ISV Competency
- Financial Services Technology Competency
- Life Sciences ISV Competency
- Machine Learning ISV Competency
- Migration and Modernization ISV Competency
- Retail ISV Competency

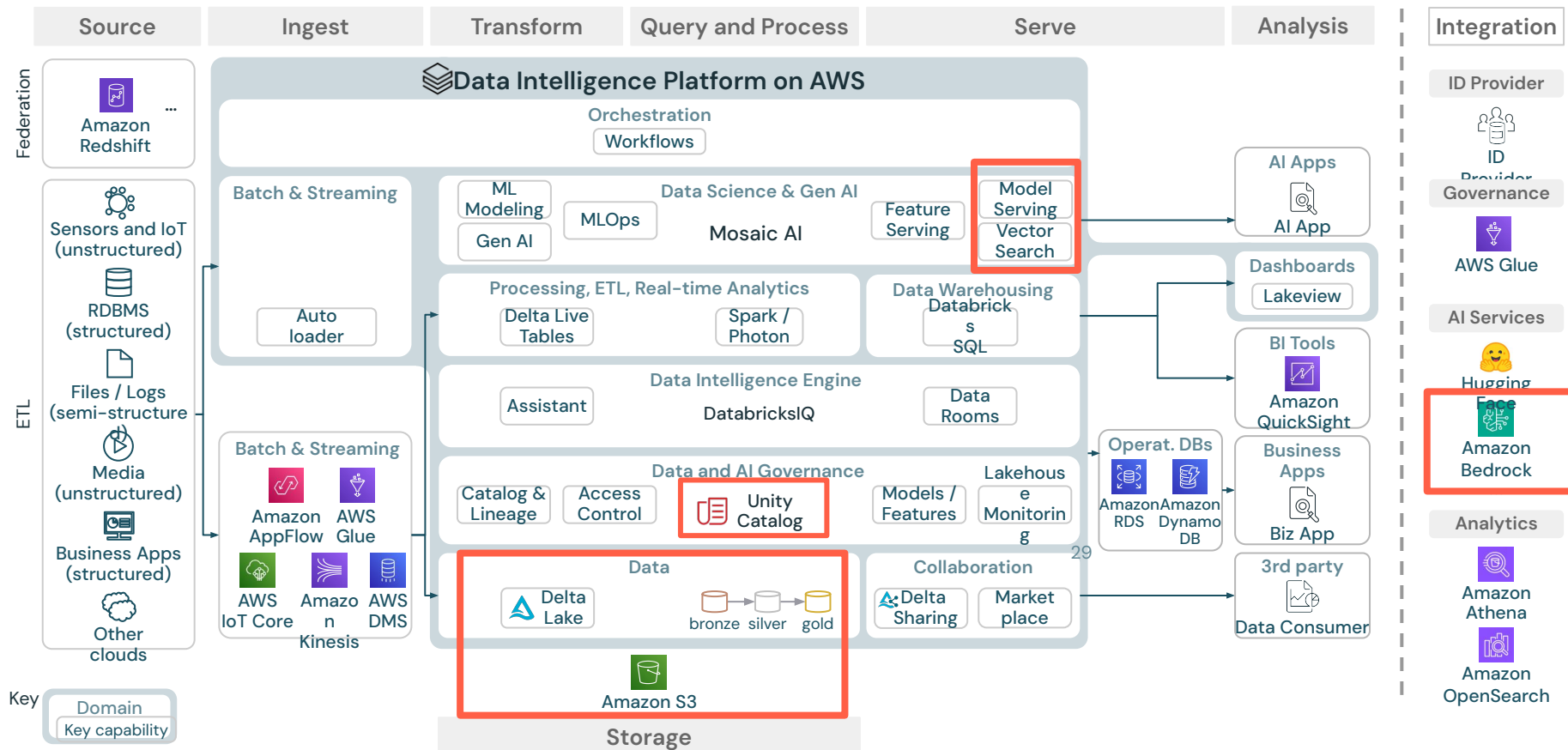
AWS Service validations:

- AWS Graviton Ready Product
- Amazon EC2 Spot Ready Product
- Amazon Redshift Ready Product

Databricks on AWS: High-level Architecture



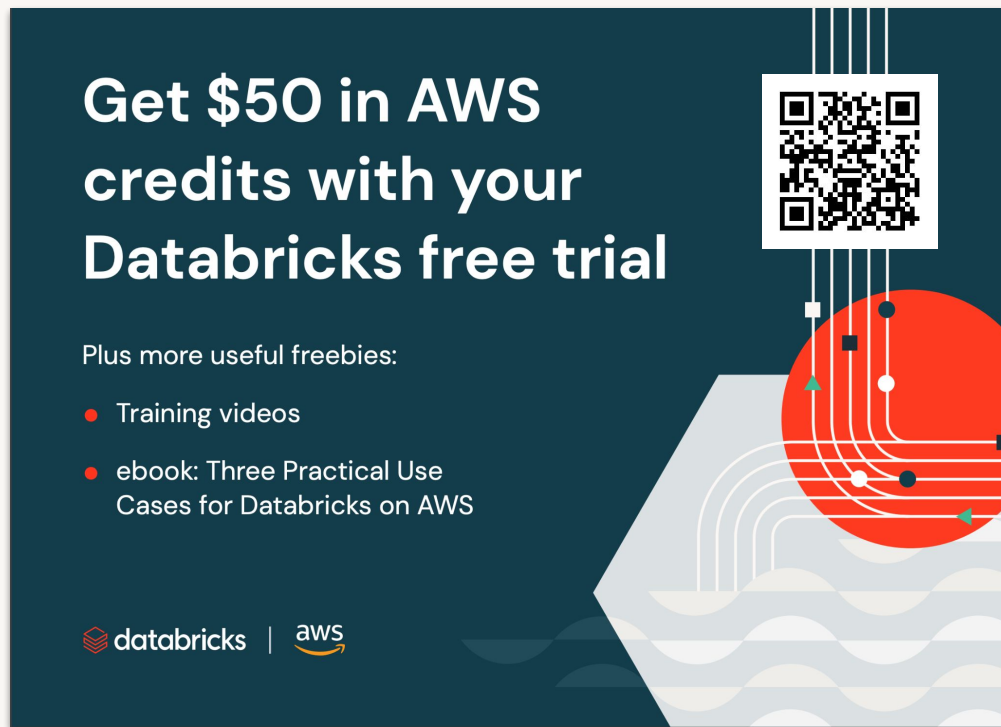
Databricks Data Intelligence Platform on AWS



Databricks on AWS Marketplace Starter Kit

- 14-day free trial
- Training videos
- eBook: Three Practical Use Cases for Databricks on AWS

Start your free trial today



Get \$50 in AWS credits with your Databricks free trial

Plus more useful freebies:

- Training videos
- ebook: Three Practical Use Cases for Databricks on AWS

 databricks |  aws



Hands-on Labs

Lab registration link: <https://bit.ly/42Ofd9c>

1. Data Engineering Hands-On Lab
2. Data Science & ML Hands-On Lab
3. Databricks SQL Hands-On Lab



Questions?



